

$$\begin{aligned}
& \frac{21}{16\pi^2} \left(-\frac{1}{9} g_2^6 \text{LF}_{3,0}[\mu_2] - \frac{1}{6} g_2^6 \text{LF}_{4,-1}[\mu_2] + \right. \\
& \frac{8}{45} g_2^6 \text{LF}_{5,-2}[\mu_2] + \frac{1}{72} (-18 g_2^2 g_2^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} + \sum_p g_2^4 (9 g_1^2 + 5 g_2^2)) \text{LF}_{3,0}[\mu_1^{\text{P}}] - \\
& \frac{1}{48} \sum_p g_2^6 \text{LF}_{4,-1}[\mu_1^{\text{P}}] - \frac{2}{45} \sum_p g_2^6 \text{LF}_{5,-2}[\mu_1^{\text{P}}] + \\
& \frac{1}{24} g_2^2 (6 g_1^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} + 6 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} (g_1^2 - 9 g_2^2) + \sum_p g_2^2 (-3 g_1^2 + 5 g_2^2)) \text{LF}_{3,0}[\mu_1^{\text{P}}] + \\
& \left(\frac{3}{2} g_2^4 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} - \frac{1}{16} \sum_p g_2^6 \right) \text{LF}_{4,-1}[\mu_1^{\text{P}}] - \frac{2}{15} \sum_p g_2^6 \text{LF}_{5,-2}[\mu_1^{\text{P}}] - \frac{1}{18} g_2^6 \text{LF}_{3,0}[\tilde{\mu}] - \\
& \frac{1}{12} g_2^6 \text{LF}_{4,-1}[\tilde{\mu}] + \frac{4}{45} g_2^6 \text{LF}_{5,-2}[\tilde{\mu}] - g_1^6 \tilde{\mu}^2 \text{LF}_{3,3,-2}[\mu_1, \tilde{\mu}] - g_1^6 m_1^2 \tilde{\mu}^2 \text{LF}_{3,3,-1}[\mu_1, \tilde{\mu}] + \\
& \frac{2}{3} g_2^6 \text{LF}_{2,1,0}[\mu_2, \tilde{\mu}] + \frac{17}{12} g_2^6 \text{LF}_{2,2,-1}[\mu_2, \tilde{\mu}] - \frac{1}{3} g_2^6 \text{LF}_{3,1,-1}[\mu_2, \tilde{\mu}] - \frac{7}{3} g_2^6 \text{LF}_{3,2,-2}[\mu_2, \tilde{\mu}] + \\
& \frac{3}{2} g_2^6 m_2^2 \text{LF}_{3,2,-1}[\mu_2, \tilde{\mu}] + 3 g_2^6 \text{LF}_{3,3,-3}[\mu_2, \tilde{\mu}] - 3 g_2^6 (m_2^2 + \tilde{\mu}^2) \text{LF}_{3,3,-2}[\mu_2, \tilde{\mu}] - \\
& 3 g_2^6 m_2^2 \tilde{\mu}^2 \text{LF}_{3,3,-1}[\mu_2, \tilde{\mu}] + g_2^6 \text{LF}_{4,2,-3}[\mu_2, \tilde{\mu}] - g_2^6 m_2^2 \text{LF}_{4,2,-2}[\mu_2, \tilde{\mu}] + \\
& \frac{1}{2} g_1^2 g_2^2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{2,2,0}[\mathbf{m}_d^{\text{r}}, \mathbf{m}_q^{\text{P}}] - \frac{1}{2} g_1^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \overline{y_u^{\text{st}}} y_u^{\text{st}} \text{LF}_{2,1,0}[\mathbf{m}_d^{\text{r}}, \mathbf{m}_u^{\text{t}}] + \\
& \frac{1}{2} g_1^2 g_2^2 \overline{a_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{2,2,0}[\mathbf{m}_e^{\text{r}}, \mathbf{m}_l^{\text{P}}] - \frac{1}{4} g_1^2 g_2^2 (\overline{a_e^{\text{pr}}} a_e^{\text{pr}} + \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \text{LF}_{3,2,-1}[\mathbf{m}_l^{\text{P}}, \mathbf{m}_e^{\text{r}}] + \\
& \frac{3}{4} g_1^2 g_2^2 (-\overline{a_e^{\text{pr}}} a_e^{\text{pr}} + \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \text{LF}_{4,1,-1}[\mathbf{m}_l^{\text{P}}, \mathbf{m}_e^{\text{r}}] - \\
& \frac{1}{6} g_2^2 (-3 g_1^2 \overline{a_e^{\text{pr}}} a_e^{\text{pr}} + \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} (3 g_1^2 + g_2^2)) \text{LF}_{5,1,-2}[\mathbf{m}_l^{\text{P}}, \mathbf{m}_e^{\text{r}}] + \\
& 2 g_1^2 g_2^2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_q^{\text{P}}, \mathbf{m}_d^{\text{r}}] - \frac{1}{4} g_1^2 g_2^2 (\overline{a_d^{\text{pr}}} a_d^{\text{pr}} + \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \text{LF}_{3,2,-1}[\mathbf{m}_q^{\text{P}}, \mathbf{m}_d^{\text{r}}] + \\
& \frac{3}{4} g_1^2 g_2^2 (-5 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} + \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \text{LF}_{4,1,-1}[\mathbf{m}_q^{\text{P}}, \mathbf{m}_d^{\text{r}}] - \\
& \frac{1}{2} g_2^2 (-3 g_1^2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} + \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} (3 g_1^2 + g_2^2)) \text{LF}_{5,1,-2}[\mathbf{m}_q^{\text{P}}, \mathbf{m}_d^{\text{r}}] - \\
& \frac{1}{4} \overline{y_u^{\text{st}}} y_u^{\text{pt}} (\overline{y_d^{\text{pr}}} y_d^{\text{sr}} (g_1^2 - 3 g_2^2) - 12 g_2^2 \overline{y_u^{\text{pr}}} y_u^{\text{sr}}) \text{LF}_{2,1,0}[\mathbf{m}_q^{\text{P}}, \mathbf{m}_q^{\text{S}}] - \\
& \frac{3}{2} g_2^2 \overline{y_u^{\text{pr}}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \text{LF}_{3,1,-1}[\mathbf{m}_q^{\text{P}}, \mathbf{m}_q^{\text{S}}] + \\
& \frac{1}{4} g_2^2 (-\overline{a_u^{\text{pr}}} a_u^{\text{pr}} (3 g_1^2 + 5 g_2^2) + g_1^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{2,2,0}[\mathbf{m}_q^{\text{P}}, \mathbf{m}_u^{\text{r}}] + \\
& \frac{1}{4} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} (g_1^2 - 9 g_2^2) - g_1^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{3,1,0}[\mathbf{m}_q^{\text{P}}, \mathbf{m}_u^{\text{r}}] + \\
& g_2^4 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_q^{\text{P}}, \mathbf{m}_u^{\text{r}}] + \frac{3}{2} g_2^4 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_q^{\text{P}}, \mathbf{m}_u^{\text{r}}] - \\
& \frac{1}{4} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} (g_1^2 - 3 g_2^2) \text{LF}_{2,1,0}[\mathbf{m}_q^{\text{S}}, \mathbf{m}_q^{\text{P}}] - \\
& \frac{1}{4} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} (7 g_1^2 + g_2^2) + g_1^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{3,1,0}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{P}}] + \\
& \frac{1}{4} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} (g_1^2 + 5 g_2^2) + g_1^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{3,2,-1}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{P}}] + \\
& \frac{3}{4} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} (5 g_1^2 + g_2^2) - g_1^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{4,1,-1}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{P}}] - \\
& \frac{1}{2} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} (3 g_1^2 + g_2^2) - 3 g_1^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{5,1,-2}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{P}}] + \\
& \frac{1}{2} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} (2 g_1^2 + 3 g_2^2) \text{LF}_{2,1,0}[\mathbf{m}_u^{\text{t}}, \mathbf{m}_d^{\text{r}}] - \\
& \frac{3}{2} g_2^2 \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{3,1,-1}[\mathbf{m}_u^{\text{t}}, \mathbf{m}_d^{\text{r}}] + \frac{1}{2} g_1^2 g_2^4 \text{LF}_{3,1,-1}[\tilde{\mu}, m_1] - \\
& \frac{3}{4} g_1^4 g_2^2 \text{LF}_{3,2,-2}[\tilde{\mu}, m_1] - \frac{3}{4} g_1^4 g_2^2 m_1^2 \text{LF}_{3,2,-1}[\tilde{\mu}, m_1] - \frac{7}{12} g_1^2 g_2^4 \text{LF}_{4,1,-2}[\tilde{\mu}, m_1] + \\
& \frac{1}{2} g_1^4 g_2^2 \text{LF}_{4,2,-3}[\tilde{\mu}, m_1] + \frac{1}{2} g_1^4 g_2^2 m_1^2 \text{LF}_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{6} g_1^2 g_2^4 \text{LF}_{5,1,-3}[\tilde{\mu}, m_1] + \\
& \frac{1}{6} g_2^6 \text{LF}_{3,1,-1}[\tilde{\mu}, m_2] - \frac{65}{12} g_2^6 \text{LF}_{3,2,-2}[\tilde{\mu}, m_2] - \frac{3}{4} g_2^6 m_2^2 \text{LF}_{3,2,-1}[\tilde{\mu}, m_2] - \frac{3}{4} g_2^6 \text{LF}_{4,1,-2}[\tilde{\mu}, m_2] \\
& 3 g_2^6 \text{LF}_{4,2,-3}[\tilde{\mu}, m_2] + \frac{1}{2} g_2^6 (m_2^2 + \tilde{\mu}^2) \text{LF}_{4,2,-2}[\tilde{\mu}, m_2] + \frac{1}{2} g_2^6 \text{LF}_{5,1,-3}[\tilde{\mu}, m_2] - \\
& 2 g_1^2 g_2^4 \text{LF}_{2,2,1,-2$$